$$\frac{d}{dt}\langle \mathbf{L} \rangle = -\frac{i}{\hbar}\langle [\mathbf{L}, \mathbf{H}] \rangle. \qquad (4.18)$$

This is a very interesting and important equation. It relates the time derivative of the expectation value of an observable **L** to the expectation value of another observable, namely– $\frac{i}{\hbar}$ [**L**, **H**].

> **Exercise 4.2:** Prove that if **M** and **L** are both Hermitian, *i*[**M**, **L**] is also Hermitian. Note that the *i* is important. The commutator is, by itself, not Hermitian.

If we assume that the probabilities are nice, narrow, bell-shaped curves, then Eq. 4.18 tells us how the peaks of the curves move with time. Equations like this are the closest thing in quantum mechanics to the equations of classical physics. Sometimes we even omit the angle brackets in such equations and write them in a shorthand form:

$$\frac{d\mathbf{L}}{dt} = -\frac{i}{\hbar}[\mathbf{L}, \mathbf{H}]. \qquad (4.19)$$

But keep in mind that a quantum equation of this type should be in the middle of a sandwich, with a bra <Ψ| on one side, and a ket |Ψ> on the other. Alternatively, we can think of it as an equation that tells us how the centers of probability distributions move around.

Does Eq. 4.19 have a familiar look to it? If not, go back to Lectures 9 and 10 in *Volume I*, where we learned about the Poisson bracket formulation of classical mechanics. On page 172, the following equation can be found:[3]

$$\dot{F} = \{F, H\}. \qquad (4.20)$$